vesuvius-challenge-surface-detection

12th place solution

Rank	12
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Source: https://www.kaggle.com/competitions/vesuvius-challenge-surface-detection/discussion/679259
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Thank you for the organizers and Kaggle team for this great competition!

1. Summary

nnUNet-based 3D segmentation. 4-model ensemble of 3d_lowres and 3d_fullres + opening/closing post-processing

- Public LB: 0.578
- Private LB: 0.613

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2. Solution Overview

```
[Pipeline]

Input (3D CT volume)
    ↓
nnUNet preprocessing (normalization, resampling)
    ↓
4-model inference (2x T4 GPU parallel)
  ├─ 3d_lowres fold_0 (2000ep) × 0.2
  ├─ 3d_lowres fold_1 (4000ep) × 0.2
  ├─ 3d_fullres fold_0 (4000ep) × 0.3
  └─ 3d_fullres fold_1 (2000ep) × 0.3
    ↓
Weighted average of probability maps
    ↓
Post-processing
  ├─ Hysteresis thresholding (t_low=0.3, t_high=0.85)
  ├─ Opening (noise removal)
  └─ Closing (hole filling)
    ↓
Output (3D segmentation mask)
```

Key Points

- Used nnUNet with mostly default settings
- Combined lowres + fullres for different scales
- Post-processing significantly improved TopoScore (topology quality)

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3. Model & Training

Configuration Comparison

	3d_lowres	3d_fullres
Patch size	128^3	128^3
Spacing	1.56 mm	1.0 mm
Effective FOV	~200 mm^3	~128 mm^3

	3d_lowres	3d_fullres
Image size	205^3	320^3

*Same patch size but different physical coverage due to spacing differences

Fold Strategy

- Used only 2 folds from nnUNet's default 5-fold CV (fold_0, fold_1)
- Reason: Time constraints, and 2 folds provided sufficient diversity
- Training data: 628 cases, validation data: 157-158 cases/fold

Epochs

- Base: 2000 epochs
- Some models extended to 4000 epochs
- Wanted to train all models to 4000 epochs but ran out of competition deadline
- Final 4 models used:

Config	Fold	Epochs	CV (opening_closing)
3d_lowres	fold_0	2000	0.606
3d_lowres	fold_1	4000	0.603
3d_fullres	fold_0	4000	0.606
3d_fullres	fold_1	2000	0.603

Effect of Epochs

- 1000ep → 2000ep:
  - 3d_lowres fold_0: 0.601 → 0.605 (+0.004)
- 2000ep → 4000ep:
  - 3d_fullres fold_0: 0.603 → 0.606 (+0.003)
  - 3d_lowres fold_1: 0.594 → 0.603 (+0.009)

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4. Inference & Post-processing

Sliding Window Settings

- step_size = 0.3 (70% overlap)

TTA (Test Time Augmentation)

- 8-direction mirroring
- Used TTA for inference, switched to non-TTA when approaching 9-hour time limit

Post-processing Pipeline

```
Probability map → Hysteresis → Opening → Closing → Dust Removal → Final mask
```

Step 1: 3D Hysteresis Thresholding

- t_high = 0.85: High confidence regions (seeds)
- t_low = 0.30: Wide coverage
- Structuring element: generate_binary_structure(3, 3) - 26-connectivity
- Process: binary_propagation from strong to weak regions

Step 2: Opening (Noise Removal)

- Structuring element: generate_binary_structure(3, 1) - 6-connectivity
- Effect: Removes small protrusions and noise

Step 3: Anisotropic Closing (Hole Filling)

- Structuring element: z_radius=2, xy_radius=1 (anisotropic)
- Larger z-direction to strengthen inter-slice connectivity

Step 4: Dust Removal

- min_size = 100 voxels
- Effect: Removes small isolated regions

Post-processing Effect Comparison (2000ep fold_0)

Method	Competition Score	TopoScore	Improvement
None (argmax)	0.571	0.246	-
Hysteresis only	0.592	0.322	+0.021
+ Opening/Closing	0.606	0.342	+0.035

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5. Scores

Score Progression

Stage	Changes	Public	Private
Baseline	1000ep, 1 model, argmax	0.530	0.552
+Post-processing	hysteresis	0.549	0.570
+2-fold	fold_0+1 ensemble	0.565	0.590
+opening_closing	Improved post-processing	0.565	0.593
+2000ep	Increased epochs	0.568	0.593
+fullres	fullres 2 models only	0.575	0.598

Stage	Changes	Public	Private
+4 models	lowres+fullres ensemble	0.582	0.606
+Weight tuning	fullres 60%, lowres 40%	0.583	0.605
+TTA	8-direction mirroring	0.584	0.607
+4000ep	Final submission	0.578	0.613

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6. What Worked / What Didn't Work

What Worked

- nnUNet default settings
- 4-model ensemble (lowres + fullres fold0, 1)
- Post-processing (Hysteresis + Opening/Closing)
- Epoch increase (1000→2000→4000)
- TTA (Test Time Augmentation)

What Didn't Work

- Increased inference parallelism (npp=2, nps=2): Counterproductive due to T4 memory constraints (+50min slower)
- step_size changes: 0.3 was best, 0.2 and 0.5 worsened Public score